BAHIRDAR UNIVERSTY

BAHIRDAR INSTITUTE OF TECHNOLOGY

Faculty of computing Department of Software Engineering

Course: operating system programing project

Title:**LOCUS OS**

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1. Introduction

1.1 Background

In this project, I studied the LOCUS operating system, which is a distributed operating system developed in the early 1980s at the University of California, Los Angeles (UCLA). LOCUS was designed to allow multiple computers connected through a network to work together as a single system by sharing resources such as files and processing power.

From my research, I understood that LOCUS is very different from modern operating systems like Windows or Linux. It was mainly developed for academic and research purposes and was designed to run on older hardware systems such as PDP-11 and VAX computers. These systems are now outdated and no longer used in modern computing environments.

I also found that LOCUS OS does not provide modern installation features such as ISO files or official download platforms. Because of this, it is difficult to install or run LOCUS OS on current computers or virtual machines

1.2 Motivation

The main motivation for this project is to improve my understanding of operating systems, especially distributed systems like LOCUS. While working on this project, I became interested in how older systems were designed and how they contributed to the development of modern operating systems.

Even though I was not able to install LOCUS OS practically due to its limitations, I learned a lot by researching and analyzing the system. This helped me understand the challenges of working with outdated technologies and improved my problem-solving skills.

In addition, this project helped me understand the concept of virtualization and the difficulties involved in running old operating systems in modern environments. Overall, this experience has increased my knowledge and interest in operating systems and system programming.

2. Project Objectives

The main objective of this project is to understand the concepts of operating systems and apply them in a practical way using a virtual environment. In this project, I focused on the LOCUS operating system, which is a historical distributed system developed for academic and research purposes. The aim was to connect theoretical knowledge with practical understanding through virtualization and system-level study.

✧ Preparing Virtual Environment

I prepared a virtual environment using Oracle VM VirtualBox to simulate a real computer system. This helped me understand how a physical computer can run multiple operating systems inside it using virtualization technology. I also learned how system resources such as RAM, CPU, and storage are allocated for virtual machines.

✧ Understanding LOCUS Operating System

I studied the LOCUS operating system to understand its structure, design, and working principles. I focused on how it functions as a distributed operating system and how it allows multiple systems to work together as a single computing environment.

✧ Studying Distributed System Concepts

To learned about distributed systems and how they enable multiple computers to share resources, communicate, and operate as one unified system. This helped me understand important concepts like resource sharing, network communication, and system coordinattio

✧ Analyzing Installation Challenges

investigated the difficulties of installing LOCUS OS on modern systems. I found that it cannot be installed because it does not provide ISO files, lacks official support, and is not compatible with current hardware and virtualization tools. I also observed errors during boot attempts, which helped me understand real installation limitations.

✧ Understanding Virtualization

I learned how virtualization works and how tools like VirtualBox allow users to run different operating systems inside a single machine. This concept helped me understand how testing and development environments are created safely without affecting the main system.

✧ **Improving System Knowledge**

Through this project, I improved my understanding of operating systems, virtualization, and system programming. I also gained practical knowledge of how operating systems function internally and how system-level operations are handled.

Overall Objective

Overall, I aimed to develop a clear understanding of both theoretical and practical aspects of operating systems. This project helped me learn about virtualization, distributed systems, system calls, and the limitations of legacy operating systems in modern computing environments.

3. Requirements

In this section, I describe the hardware and software requirements needed to prepare the virtual environment and attempt the installation of the LOCUS operating system. Since LOCUS OS does not provide any installation files or official support, the requirements are focused on creating a virtual machine to test and demonstrate these limitations.

3.1 Hardware Requirements

Processor (CPU):Originally, LOCUS OS was designed to run on older hardware systems such as PDP-11 and VAX machines, which had very low processing power(in MHz). In my setup, I used a modern processor (Intel/AMD) with virtualization support.

Memory (RAM): used 4 GB RAM for the virtual machine, which is far more than required, but still LOCUS cannot be installed due to lack of support.

Storage (Hard Disk): The original LOCUS system used very small storage capacity, usually between 5 MB to 100 MB. In the virtual environment, Iallocated around 20 GB storage

3.2 Software Requirements

Host Operating System:I used a modern operating system (Windows 10/11) to install and run VirtualBox for testing purposes**.**

Virtualization Software : Oracle VM VirtualBox was used to create a virtualmachine and attempt the installation of LOCUS OS.

LOCUS Operating System Availability:LOCUS OS does not provide any official ISO file or bootable installation media. During my research and setup, I was unable to find any downloadable installation file.

Documentation and Support
LOCUS OS has very limited documentation and no official support for installation on

modern systems. Most available information is only for academic and historical reference.

Overall Requirement

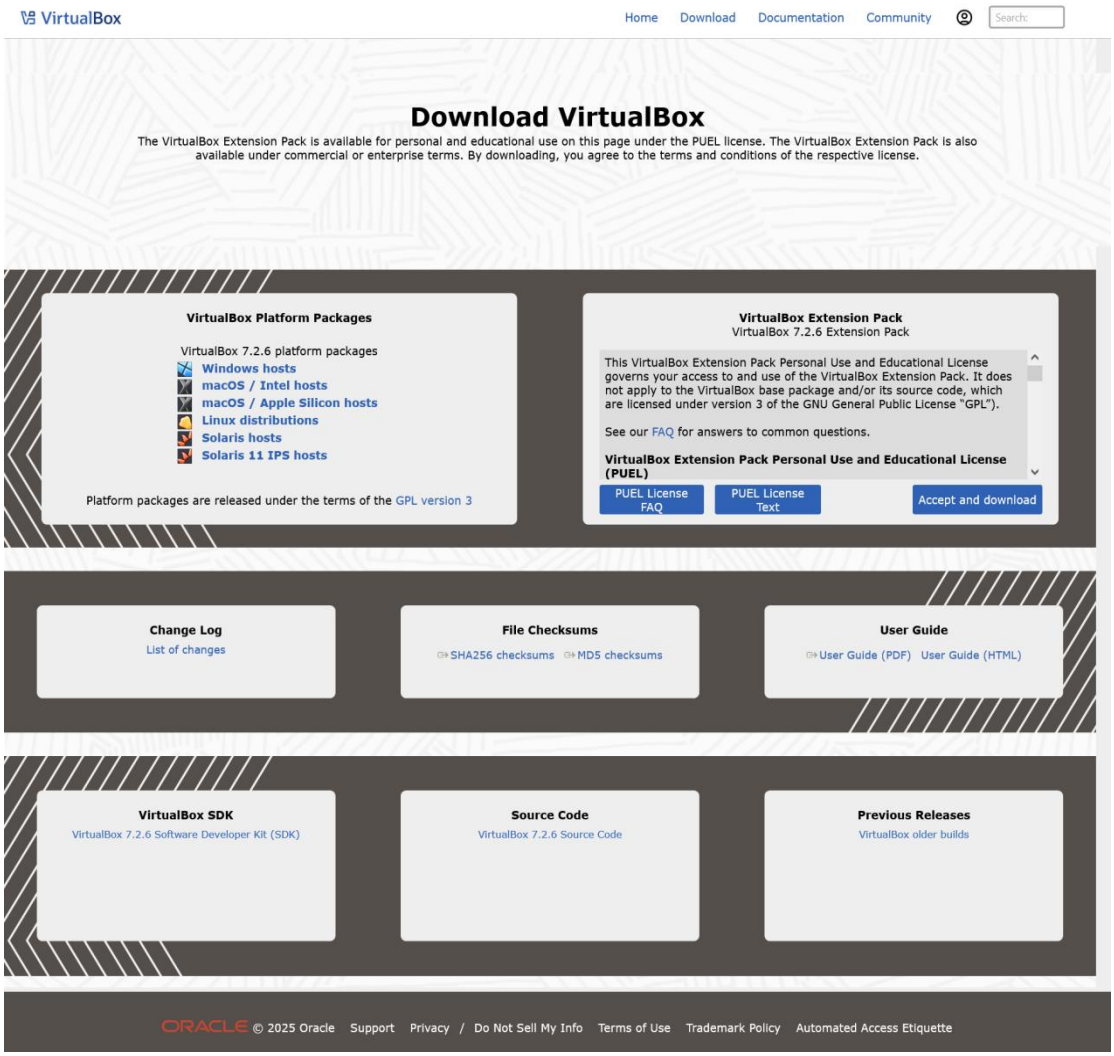
Overall, LOCUS OS requires very low hardware resources, but it cannot be installed in modern environments due to lack of ISO files, hardware incompatibility, and absence of official support. The virtual environment was used only to demonstrate and verify these limitations.

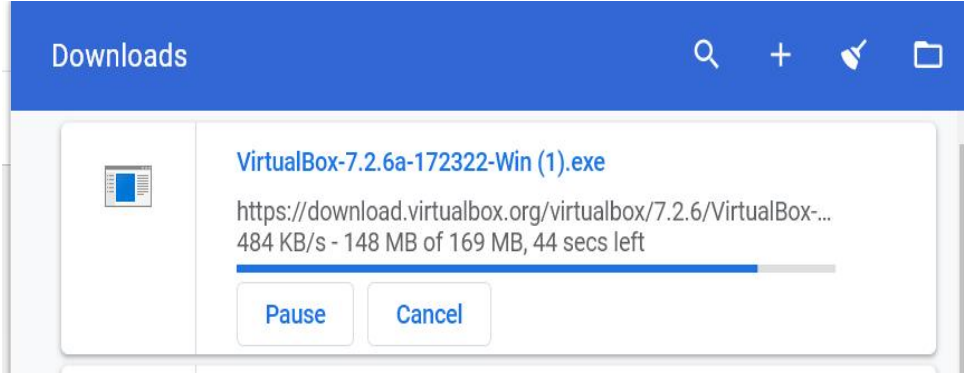
4.Instllation step

In this section, I attempted to install the LOCUS operating system using a virtual environment. Since LOCUS is a historical operating system and does not provide official installation files, the process focuses on the attempt and the issues encountered.

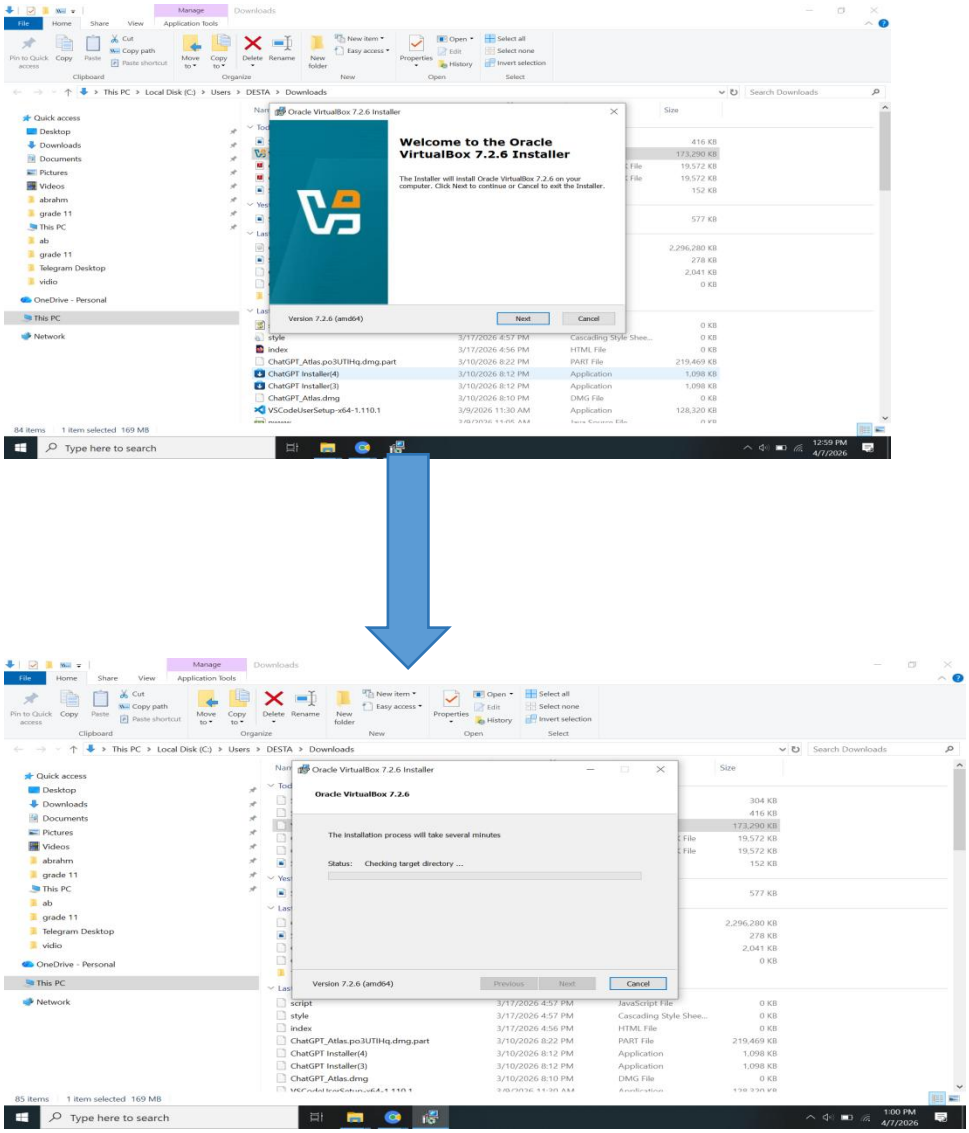
4.1 install the virtual box work station

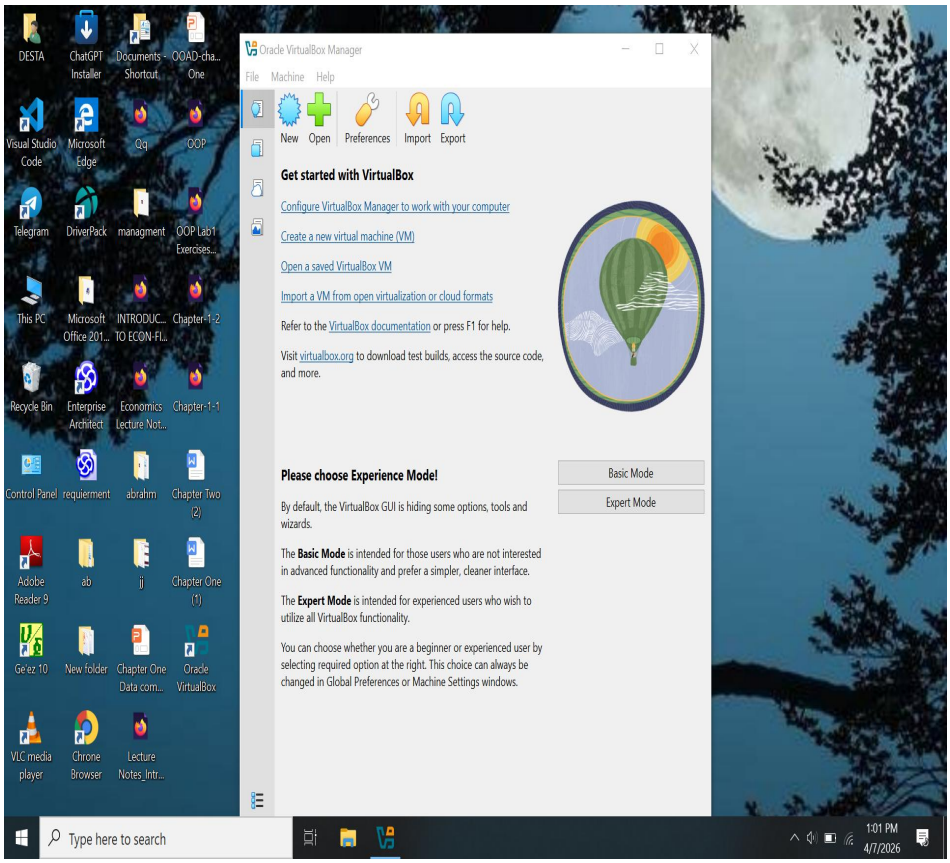
- **Brows the in website** https://www.virtualbox.org/wiki/Downloads
-





- **Virtual box Installtion step to descktop**

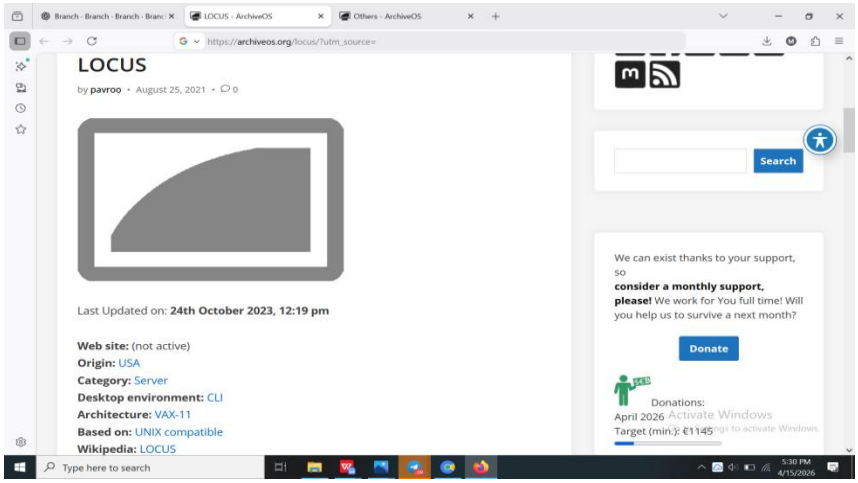


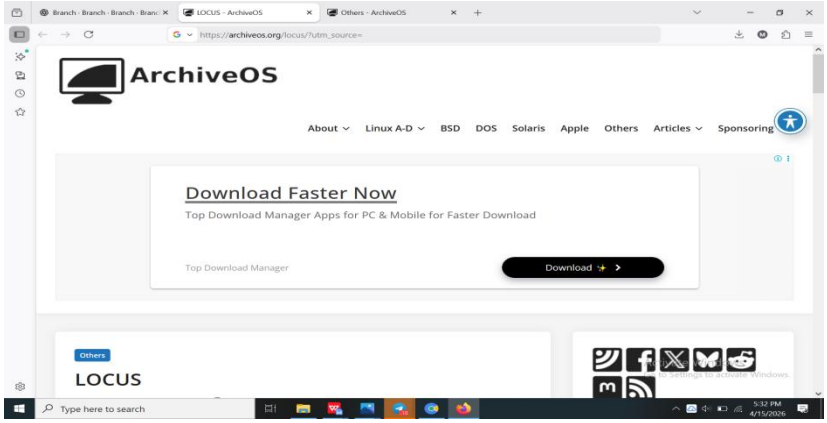


4.2 download the LOCUSE os

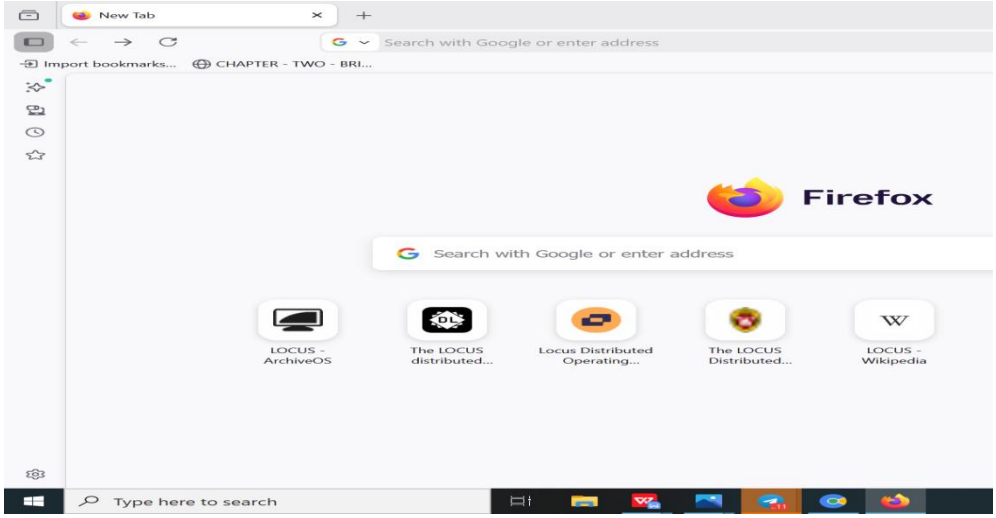
In this process I tried to install the LOCUS os system but I coudn't find any offical website even the oprating system has no any licence and \or comminty support compatable in now dayd

- **I got one web site related to the os but its not active and has no real installation instead it just justfy theorethically https://archiveos.org/locus/**





I also brows some other website as shown but I coudn't get their official website or community licence and support

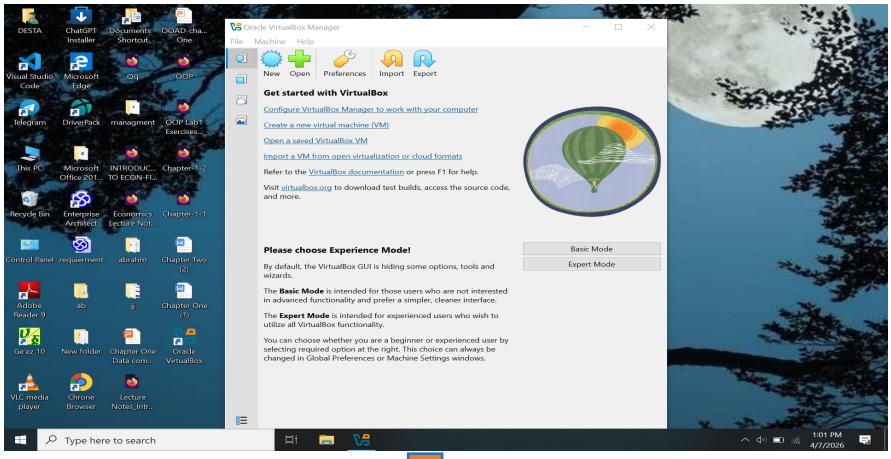


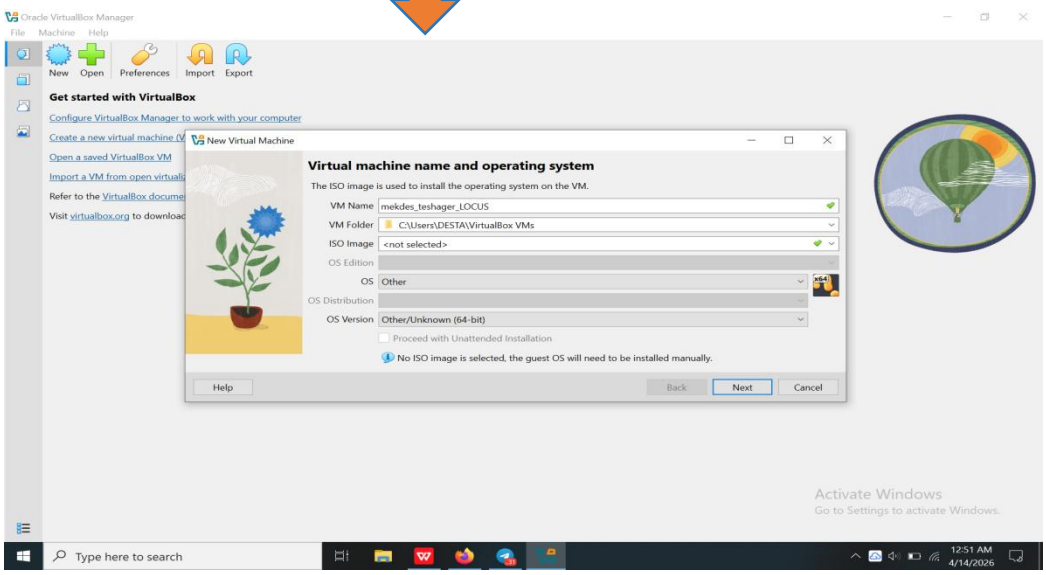
4.3 creat the virtual box

In this step ,I was just creat the virtual box that i used to compare the modern operating system and one the outdated os like LOCUS os as follow :

1 Open virtual box

✓ Click New and creat the virtual machine





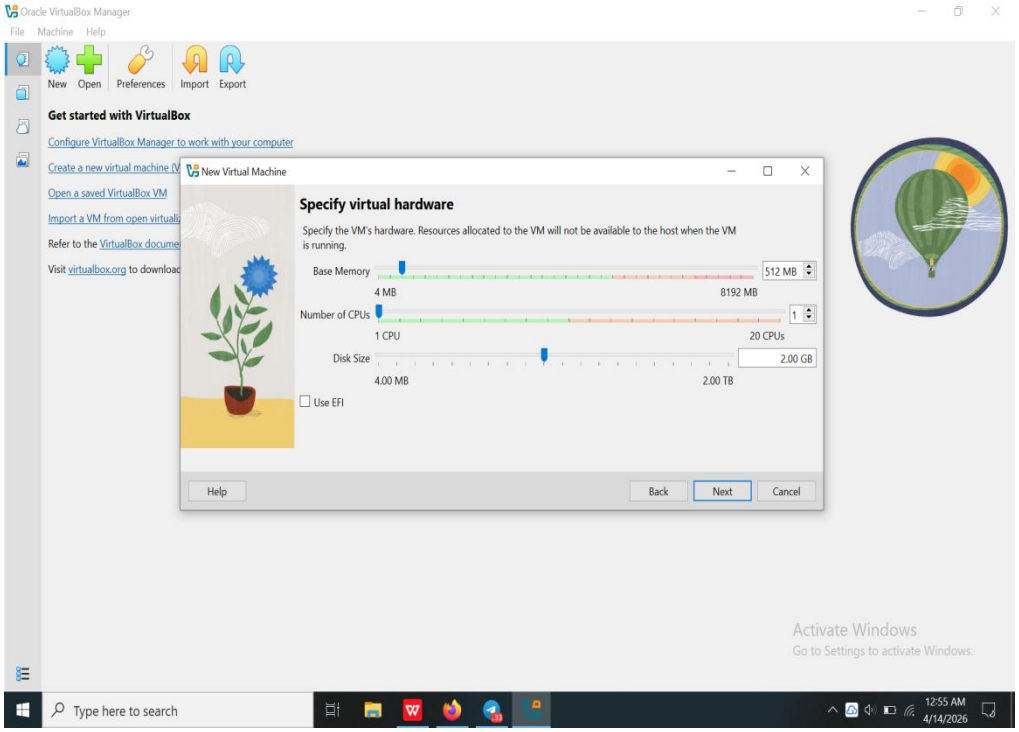
- **On this process the way I creat the machine is diffrence from other modern opersting system that different are:**

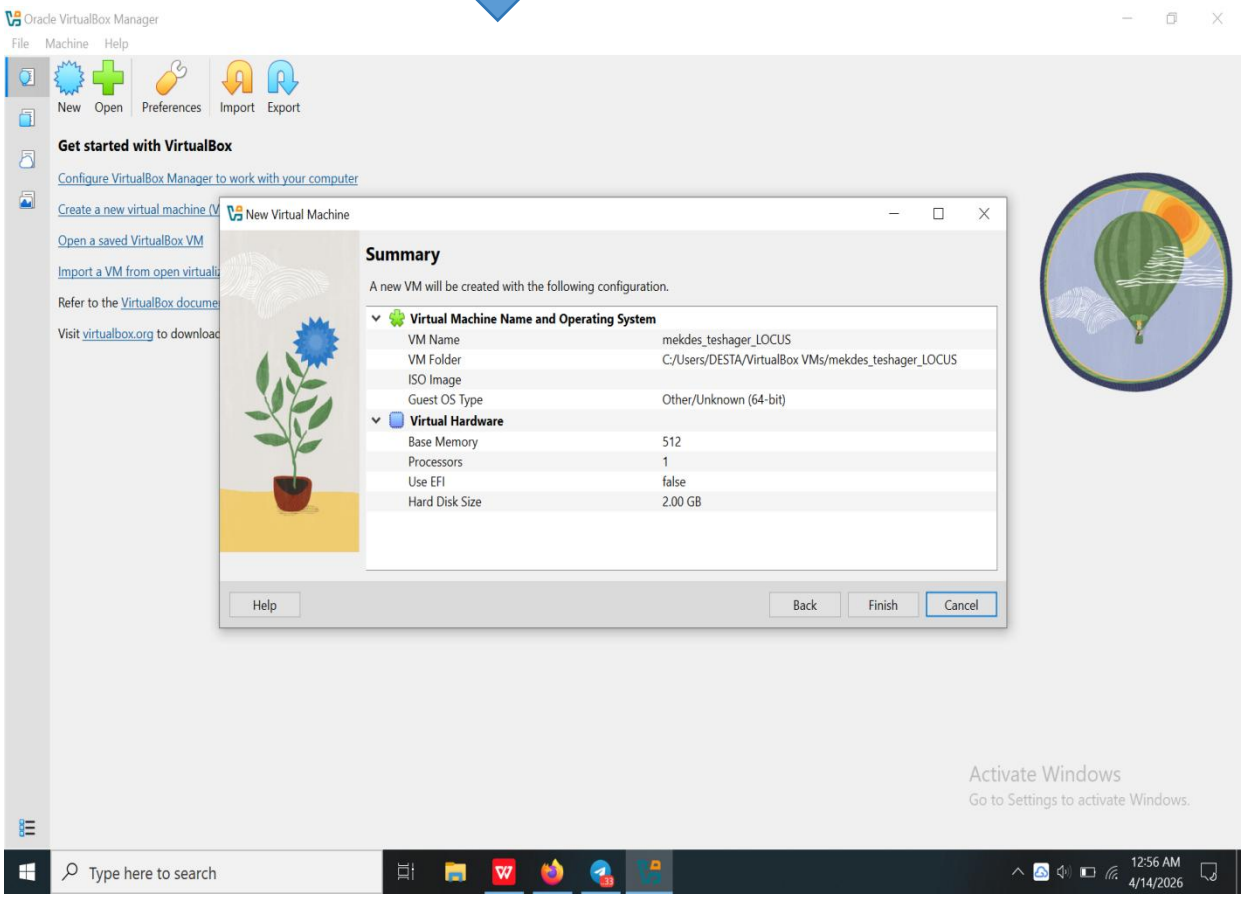
✓ the **ISO image** not select because the LOCUS has no real installtion and there is no disck

✓ The **OS** verstion **other/unkown**(64-bit) because the the LOCUS has no compatable vertion or suitable softwar

4.4.specify the virtual hard ware

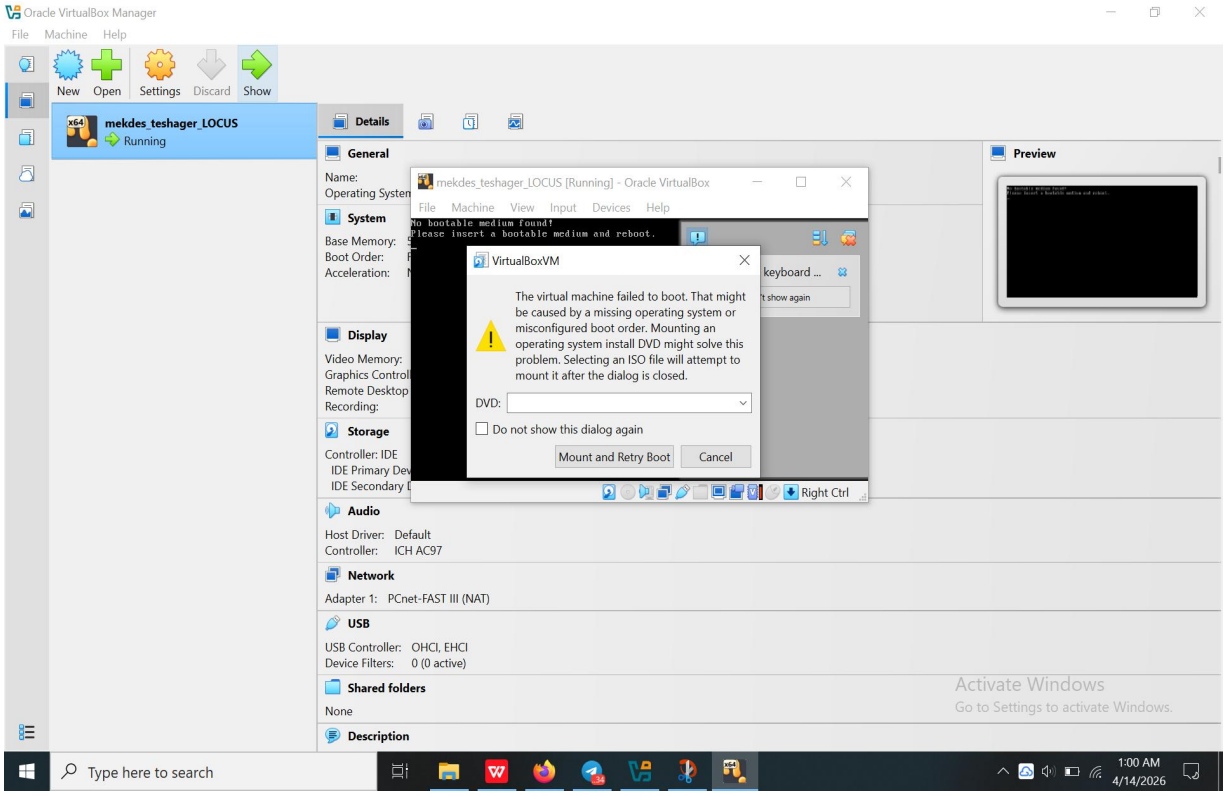
✓ **Ajust the hard ware and created accoutn as shown**


Oracle VirtualBox Manager
File Machine Help
New Open Preferences Import Export
Get started with VirtualBox
Configure VirtualBox Manager to work with your computer
Create a new virtual machine (V
Open a saved VirtualBox VM
Import a VM from open virtualiz
Refer to the VirtualBox docume
Visit virtualbox.org to downloa
New Virtual Machine
Specify virtual hardware
Specify the VM's hardware. Resources allocated to the VM will not be available to the host when the VM is running.
Base Memory
512 MB
4 MB
8192 MB
Number of CPUs
1
1 CPU
20 CPUs
Disk Size
2.00 GB
4.00 MB
2.00 TB
Use EFI
Help
Back
Next
Cancel
Activate Windows
Go to Settings to activate Windows.
Type here to search
12:55 AM
4/14/2026


Oracle VirtualBox Manager
File Machine Help
New Open Preferences Import Export
Get started with VirtualBox
Configure VirtualBox Manager to work with your computer
Create a new virtual machine (V
Open a saved VirtualBox VM
Import a VM from open virtualiz
Refer to the VirtualBox docume
Visit virtualbox.org to downloa
New Virtual Machine
Summary
A new VM will be created with the following configuration.
Virtual Machine Name and Operating System
VM Name mekdes_teshager_LOCUS
VM Folder C:/Users/DESTA/VirtualBox VMs/mekdes_teshager_LOCUS
ISO Image
Guest OS Type Other/Unknown (64-bit)
Virtual Hardware
Base Memory 512
Processors 1
Use EFI false
Hard Disk Size 2.00 GB
Help
Back
Finish
Cancel
Activate Windows
Go to Settings to activate Windows.
Type here to search
12:56 AM
4/14/2026

4.5.Try to Add ISO (install the LOCUS in vm)

✓ Click **Empty → Choose disk file but** No LOCUS OS ISO file available **gives the result as shown.**

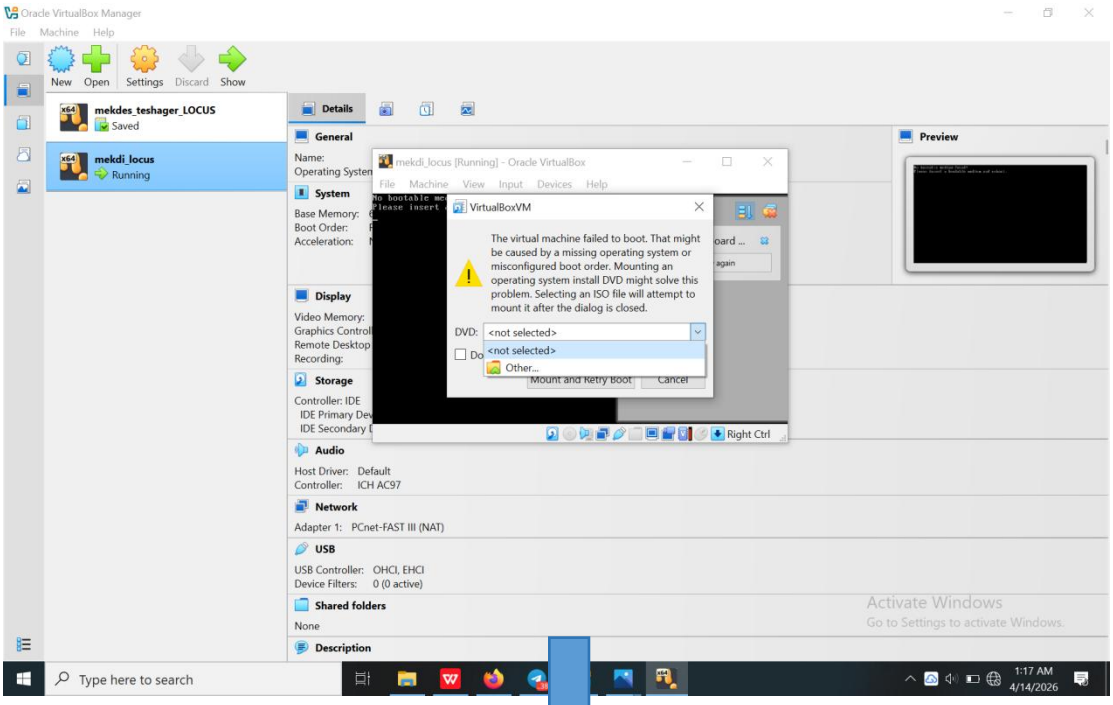


➢ **On this this stage I leav empty the DVD because there is no ISO fild in the disck gives the result 'no bootabel medium found'as shown follow**

➢

➢

➢



➢

- **So** I stopped the installation processe because

✓ Without ISO → VM cannot start installation of LOCUS

5.Major issue or problem during installation

Problem1:No Official Download

- I searched for LOCUS OS download
- But there is **no official website or installer**

Evideces:

➢ When I brows different website I only get historical and Theoretical evidence instead real installtion websit

Resorceses :

➢ These are some website shows LOCUS has no real installation web

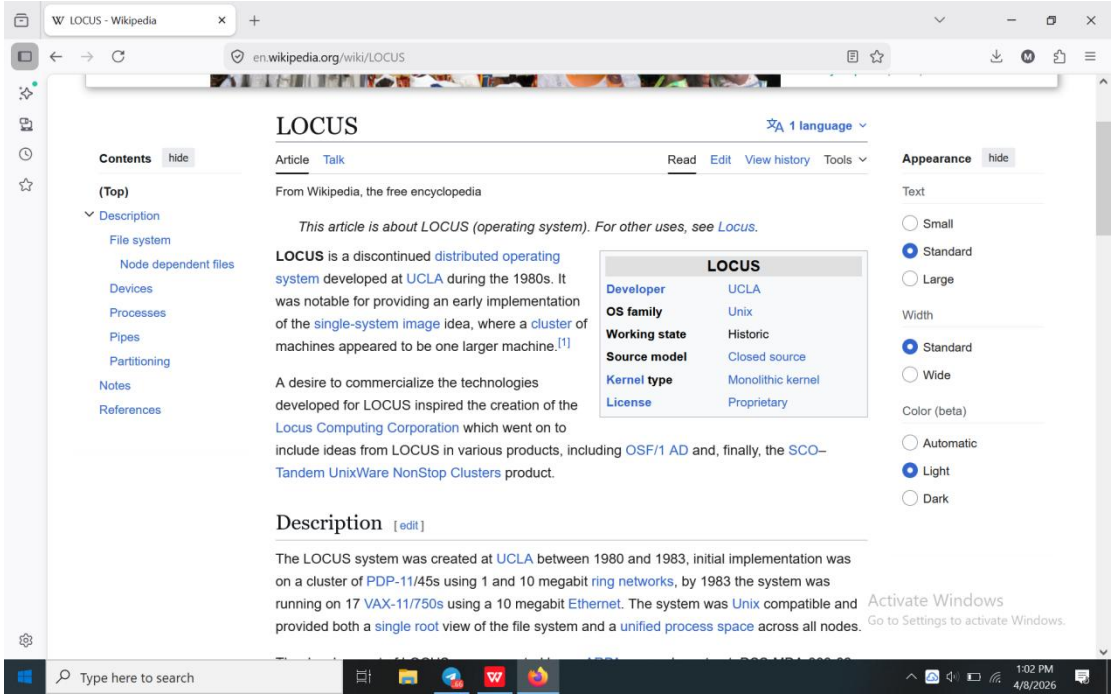
1. https://www.slideshare.net/slideshow/locus-distributed-operating-system/10961402
2. https://dl.acm.org/doi/10.1145/773379.806615
3. https://pages.cs.wisc.edu/~sschang/OS-Qual/fs/LOCUS.htm?utm_source=chatgpt.com

Those links show:

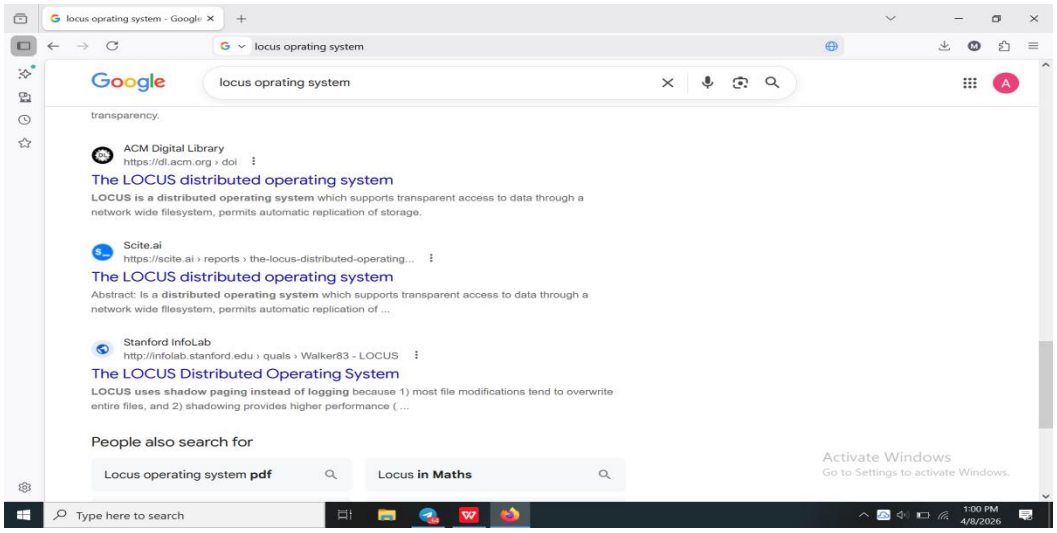
- **University (UCLA) involvement**
- **Academic nature** of LOCUS OS
- That it was a **research system**, not a consumer OS
- There is **no official download or installer** available

- Also when I search in browser directly I get theoretical evidence as shown in the screen shot

1https://en.wikipedia.org/wiki/LOCUS?utm_source=chatgpt.com



2 also when I brows only I got pdf,librarys,that explane about LOCUS thoertically as shown below



- **This is the basic and major problem that I can't do any installation process in my oprating system**

Problem 2: Problem: No ISO File

Modern OS requirement:

- Today, almost all operating systems (Windows, Linux distributions, macOS, etc.) use **ISO files** for installation.
- An ISO is a disk image that contains the full operating system and installation files.

Issue with LOCUS OS:

- LOCUS OS was a **research/distributed OS developed at UCLA in the 1980s**.
- There is **no ISO file available** for it anywhere online.
- All modern OSes expect ISO files, so without an ISO, i **cannot install LOCUS OS on a virtual machine** or physical hardware.

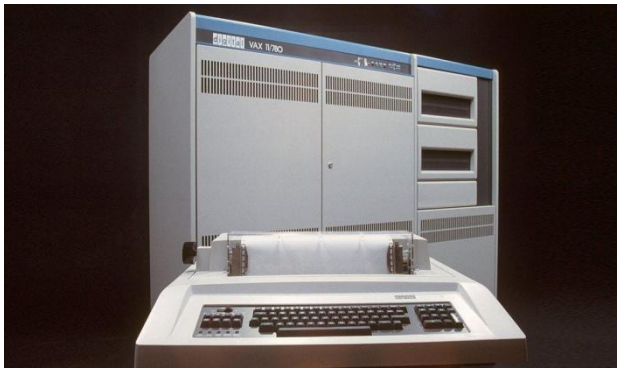
Problem 3: Hardware Incompatibility

- LOCUS runs on:
  - PDP-11
  - VAX

◆ Now days computer uses modern architecture, so it is not compatible.

IT Works as shown in the image old devices

PDP-11



VAX MINICOMPOTUR

Problem 4: Incompatibility with Virtual Machines

- Tools like VirtualBox support:
  - Windows
  - Linux
  - Modern OS

LOCUS OS:

- Needs special emulator (like SIMH → for PDP-11 ,QEMU → for many systems)
- Standard virtualization tools such as VirtualBox and VMware do not support LOCUS OS because it requires emulation of legacy hardware systems rather than modern virtual environments

Problem 5: Lack of Modern Support

- No updates
- No drivers
- No developer
- LOCUS OS is no longer maintained or supported, and therefore lacks compatibility with modern hardware drivers and system architectures.

➢ **Because of those all problem I can't do any installtion process in my LOCUS os**

6.Solotion for those problem

To address the problems encountered during the installation of LOCUS OS, I applied the following solutions and approaches:

6.1 Using Alternative Study Method

Since LOCUS OS does not have an ISO file or installation package, I used academic sources such as research papers and archive websites to study the system theoretically instead of performing a practical installation.

6.2 Using Virtualization for Demonstration

Although LOCUS OS could not be installed, I used VirtualBox to demonstrate the general process of installing a modern operating system. This helped me understand where the installation process fails in the case of LOCUS OS, especially at the ISO attachment step.

6.3 Identifying the Root Cause

I analyzed the main reasons for installation failure, including lack of ISO file, outdated hardware requirements, and incompatibility with modern virtualization tools. Understanding these causes helped me explain the problem clearly.

6.4 Considering Emulation (Advanced Solution)

I found that LOCUS OS can only run on old hardware systems such as PDP-11 and VAX. To run such systems today, specialized emulators (e.g., SIMH) are required. However, this approach is complex and not suitable for basic installation tasks.

6.5 Documentation of Problems

I carefully documented all the issues encountered during the installation attempt and provided clear explanations with supporting evidence. This ensures that the assignment is completed based on understanding rather than just practical execution.

Overall, instead of performing a direct installation, I focused on understanding, analyzing, and explaining the limitations of LOCUS OS in a modern computing environment.

7. Filesystem Support

uses its own distributed file system rather than modern file systems. It was designed to LOCUS OS is a historical distributed operating system developed in the 1980s, and it allow multiple computers to share files across a network as if they were stored on a single system.

Unlike modern operating systems, LOCUS does not support common file systems such as NTFS, FAT32, exFAT, ext4, Btrfs, ZFS, HFS+, or APFS. These file systems were developed later and are used in modern operating systems like Windows, Linux, and macOS.

The LOCUS file system is designed for distributed environments, meaning that files can be accessed from different machines transparently. This makes it different from traditional file systems, which operate on a single computer.

Comparison with Modern File Systems

- NTFS: Used in Windows for security and reliability.
- FAT32/exFAT: Used for compatibility and portable storage devices.
- ext4/Btrfs/ZFS: Used in Linux for performance, scalability, and advanced features.
- HFS+/APFS: Used in macOS for efficient storage management.

Since LOCUS OS is a distributed system, it does not use modern file systems. Instead, it uses a distributed file system designed for network-based resource sharing. Therefore, modern file systems are not applicable to LOCUS OS.

8. Advantages and Disadvantages

8.1 Advantages

Distributed System Capability
LOCUS OS allows multiple computers to work together as a single system, improving efficiency and coordination between machines.

Transparency
Users can access files and resources from different computers without knowing their exact location, making the system easier to use.

Reliability and Fault Tolerance
If one machine fails, the system can continue working using other machines, which increases reliability.

Efficient Resource Sharing
LOCUS enables sharing of files, programs, and processing power across the network, reducing redundancy and improving performance.

Scalability
New machines can be added to the system to increase performance and storage capacity.

Foundation for Modern Distributed Systems
LOCUS introduced important concepts that are used in modern distributed systems and cloud computing.

8.2 Disadvantages

Outdated Technology
LOCUS OS was developed in the 1980s and is no longer maintained or supported.

No Installation Support
It does not provide ISO files or installation packages, which makes it impossible to install on modern systems.

Hardware Incompatibility
It requires old hardware such as PDP-11 and VAX, which are not compatible with today's computers.

Limited Documentation and Resources
Most available materials are theoretical, and there are very few practical guides for installation or usage.

Complex System Design
LOCUS is more complex compared to modern operating systems, making it difficult for beginners to understand and use.

Requires Specialized Emulation
Running LOCUS today would require advanced emulators, which are difficult to configure and not practical for normal users.

LOCUS OS has strong advantages in distributed computing, but due to its outdated nature and lack of support, it is not suitable for use in modern computing environments.

Conclusion

gained a better understanding of virtualization, distributed systems, and the differences between historical and modern operating systems.In this project, I studied the LOCUS operating system and attempted to install it using a virtual environment. During this process, I learned that LOCUS OS is a historical distributed operating system developed for research purposes and designed to run on outdated hardware systems such as PDP-11 and VAX.

While following the installation steps, I encountered several challenges, including the absence of an ISO file, lack of an official download source, and incompatibility with modern virtualization tools. These issues made it impossible to perform a practical installation.

Despite these limitations, I was able to complete the project by studying the system theoretically, analyzing its architecture, and documenting the problems faced during installation. I also

Overall, this project helped me improve my knowledge, research skills, and problem-solving abilities, even though the operating system could not be installed practically.

10. Future Outlook / Recommendation

- LOCUS OS should primarily be studied from a **theoretical and historical perspective**, since it represents an early implementation of distributed operating system concepts rather than a modern, installable system.
- Due to the absence of an **official installation package, ISO file, and active support**, practical installation on modern hardware is not feasible. Therefore, hands-on implementation is highly limited.
- If practical experimentation is required, it is recommended to use specialized **legacy system emulators such as PDP-11 or VAX emulators**, which can simulate the original hardware environment. However, these tools require advanced technical knowledge and configuration.
- For students and researchers, it is more effective to use modern operating systems such as **Linux distributions (Ubuntu, Fedora, etc.)**, which provide full installation support, virtualization compatibility, and active documentation.
- LOCUS OS concepts such as **distributed file systems, process sharing, and network transparency** remain highly relevant and can be studied in relation to modern technologies like **cloud computing, distributed storage systems, and microservices architecture**.
- Future learning should emphasize the comparison between **historical operating systems and modern systems**, to understand how operating system design has evolved over time.
- It is also recommended to combine **theoretical study with virtualization practice**, using tools like VirtualBox or VMware for modern systems to gain practical experience.
- Overall, LOCUS OS serves as an important educational example that highlights the evolution of distributed systems and the limitations of early operating system designs.

2 Virtualization in Modern Operating Systems

2.1What is Virtualization?

Virtualization is a technology that allows a single physical computer system to create and run multiple virtual machines (VMs), each functioning as an independent computer with its own operating system and applications. It works by abstracting the physical hardware components such as CPU, memory, storage, and network interfaces into virtual resources.

These virtual resources are then allocated to different virtual machines, enabling them to operate independently while sharing the same physical hardware. Each virtual machine behaves as if it is a real computer, even though it is running within another system.

2.2 Why is Virtualization Used?

Virtualization is widely used in modern computing due to its numerous advantages:

- **Efficient Resource Utilization:** Multiple operating systems can run on a single physical machine, reducing hardware waste.
- **Cost Reduction:** It minimizes the need for multiple physical systems, lowering hardware and maintenance costs.
- **Isolation and Security:** Each virtual machine operates independently, so issues in one VM do not affect others.
- **Flexibility and Scalability:** New virtual machines can be easily created, modified, or removed based on requirements.
- **Testing and Development:** It allows developers to test different operating systems and applications without affecting the host system.
- **Disaster Recovery:** Virtual machines can be backed up and restored easily, improving system reliability.

2.3 How Does Virtualization Work?

Virtualization works through a special software layer called a **hypervisor** (also known as a Virtual Machine Monitor). The hypervisor is responsible for managing and distributing physical hardware resources among multiple virtual machines.

The hypervisor creates a virtual environment by:

- Allocating CPU, memory, and storage to each VM
- Managing communication between virtual machines and hardware
- Ensuring isolation between different virtual systems

There are two main types of **hypervisors**:

Type 1 (Bare-metal Hypervisor):
Runs directly on the physical hardware without a host operating system. It is more efficient and is commonly used in enterprise environments.
Examples include VMware ESXi and Microsoft Hyper-V.

Type 2 (Hosted Hypervisor):
Runs on top of an existing operating system. It is easier to use and suitable for Personal and educational purposes.
Examples include Oracle VM VirtualBox and VMware Workstation.

In this project, Oracle VM VirtualBox is used as a Type 2 hypervisor to create and manage virtual machines. However, due to the lack of installation files and hardware compatibility, LOCUS OS could not be installed in the virtual environment.

In ConclusionVirtualization is an essential technology in modern operating systems that enables efficient use of hardware resources and supports multiple operating environments. It plays a key role in cloud computing, system testing, and IT infrastructure management.

System call:accept()

The accept() system call is used in network communication to accept incoming connection requests from client systems. In the context of LOCUS OS, this system call would play an important role in enabling communication between distributed nodes in the network.

When a process on one machine wants to establish a connection, the server process uses accept() to receive and handle that connection. This supports the distributed nature of LOCUS, where multiple systems interact and share resources.

Challenges in Implementation

Due to the lack of available source code and documentation, the exact implementation details of system calls in LOCUS OS are not accessible. Additionally, since the operating system could not be installed, practical demonstration of system calls was not possible